

Experiment 5

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1 AIM

Op-Amp Applications:

- 1) Custom Weighted Summing and Difference Amplifier
- 2) Op-Amp Integrator
- 3) Precision Rectifier(Super Diode)

2 MATERIALS AND APPARATUS REQUIRED

- 1) Op-Amp (e.g., LM386, LM58)
- 2) Resistors(Carefully selected for weighting)
- 3) Capacitors
- 4) Diode(e.g., 1N4148)
- 5) Function Generator
- 6) DC Power Supply
- 7) Oscilloscope

3 THEORY

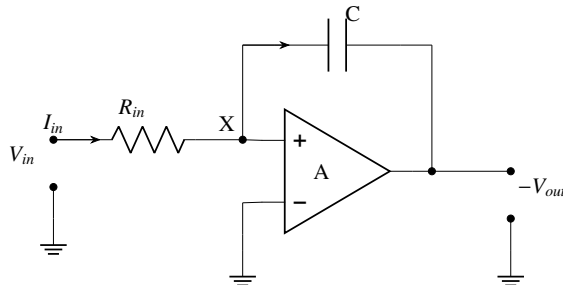
3.1 Op-Amp Integrator

Circuit Description

The basic configuration of an op-amp integrator consists of:

- An operational amplifier.
- A resistor (R_{in}) connected to the inverting input.
- A capacitor (C) connected between the output and the inverting input.
- The non-inverting input is grounded.

The circuit diagram is shown below:



Operation of the Integrator Circuit

The operation of the integrator is based on the virtual ground principle at the inverting input of the op-amp. Assuming an ideal op-amp:

- 1) The input current through R_{in} is given by:

$$I_{in} = \frac{V_{in}}{R_{in}}.$$

- 2) Since no current flows into the op-amp's inverting terminal, this current flows through the capacitor, causing it to charge or discharge.
- 3) The voltage across the capacitor (V_C) is related to the charge (Q) by:

$$Q = CV_C.$$

The current through the capacitor is:

$$I_C = C \frac{dV_C}{dt}.$$

Equating I_{in} and I_C , we have:

$$\frac{V_{in}}{R_{in}} = C \frac{dV_{out}}{dt}.$$

Rearranging and integrating both sides with respect to time gives:

$$V_{out} = -\frac{1}{R_{in}C} \int V_{in} dt,$$

where the negative sign indicates that the output is inverted.

Thus, the output voltage (V_{out}) is proportional to the time integral of the input voltage (V_{in}).

3.2 Half and Full-Wave Rectifiers

Half-Wave Precision Rectifier

The half-wave precision rectifier allows only one half of the input AC signal (positive or negative) to pass through while blocking the other half. This is achieved using an op-amp and a diode configured such that the op-amp compensates for the diode's forward voltage drop.

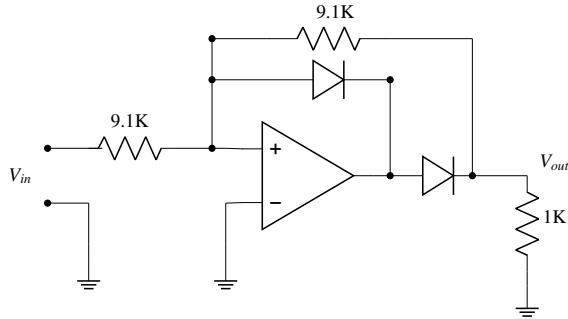
Circuit Description:

- 1) An operational amplifier (e.g., LM358).
- 2) A diode (e.g., 1N4148) connected in feedback.
- 3) Resistors to set the input and feedback paths.

Working Principle: When the input signal (V_{in}) is positive, the op-amp output drives the diode into conduction, and the output (V_{out}) follows V_{in} . When V_{in} is negative, the diode becomes reverse-biased, blocking current flow, and V_{out} remains at zero.

Mathematically:

$$V_{out} = \begin{cases} V_{in}, & V_{in} > 0, \\ 0, & V_{in} \leq 0. \end{cases}$$



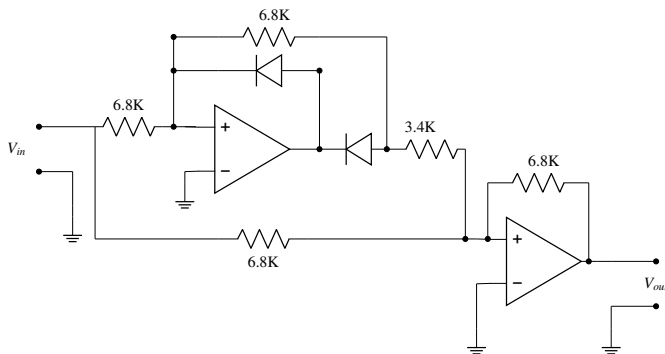
This results in a rectified waveform where only the positive half-cycles are present at the output.

Full-Wave Precision Rectifier

The full-wave precision rectifier allows both halves of the input AC signal to be rectified, producing a unipolar output. This is achieved by combining a half-wave rectifier with a summing amplifier.

Circuit Description:

- 1) A half-wave precision rectifier circuit.
- 2) A summing amplifier to combine signals from both positive and negative cycles.
- 3) Additional diodes and resistors for proper signal routing.



Working Principle:

- 1) During the positive half-cycle of V_{in} , the first op-amp operates as a half-wave rectifier, passing the positive signal to the summing amplifier.
- 2) During the negative half-cycle of V_{in} , another path inverts and shifts the signal such that it also appears as a positive contribution at the summing amplifier input. The summing amplifier combines these contributions to produce a continuous positive output for both halves of the input signal.

Mathematically: For an input sinusoidal signal $V_{in} = V_m \sin(\omega t)$, the output voltage is:

$$V_{out} = |V_{in}| = V_m |\sin(\omega t)|.$$

This results in a fully rectified waveform at the output.

4 PROCEDURE

4.1 Op-Amp Integrator

- 1) Set Up the Circuit:
 - Connect the circuit as per the integrator circuit diagram.
 - Use a resistor (R_{in}) at the input and a capacitor (C_f) in the feedback loop of the op-amp.
 - Ground the non-inverting terminal of the op-amp.
 - Connect the power supply to the op-amp (12V and -12V).
- 2) Apply Input Signal:
 - Connect a function generator to provide a square wave input signal to the circuit.
- 3) Observe Output Waveform:
 - Connect the input and output signals to two channels of a CRO.
 - Set the CRO to AC coupling mode.
 - Observe and record both input and output waveforms simultaneously.
- 4) Analyze Output Waveform:
 - For a square wave input, verify that the output is a triangular waveform.
 - Note down the amplitude and time period of both input and output signals.

4.2 Precision Rectifiers

- 1) Set Up the Circuit:
 - Connect the circuit as per the precision rectifier circuit diagram.
 - For the half-wave rectifier:
 - Use an op-amp with a diode in the feedback loop and a resistor at the input.
 - Ground the non-inverting terminal of the op-amp.
 - For the full-wave rectifier:
 - Combine two op-amps: one for inverting and rectifying the negative half-cycle and another for summing both positive and inverted signals.
 - Use appropriate diodes and resistors for signal routing.
 - Connect the power supply to the op-amp (+12V and -12V).
- 2) Apply Input Signal:
 - Connect a function generator to provide a sine wave input signal to the circuit.
- 3) Observe Output Waveform:
 - Connect the input and output signals to two channels of a CRO.
 - Set the CRO to AC coupling mode.
 - Observe and record both input and output waveforms simultaneously.

4) Analyze Output Waveform:

- For the half-wave rectifier:
 - Verify that only one half (positive or negative) of the sine wave is present at the output while the other half is blocked.
 - Note down the amplitude and time period of both input and output signals.
- For the full-wave rectifier:
 - Verify that both halves of the sine wave are rectified, resulting in a unipolar output waveform.
 - Note down the amplitude and time period of both input and output signals.

5 OBSERVATIONS

5.1 Op-Amp Integrator

6 PRECAUTIONS

- 1) Make sure not to accidentally discharge the capacitor
- 2) Make sure the DC generator is connected to ground properly
- 3) Use a push button for shorting the inductor and capacitor
- 4) Repeat the experiment multiple times to be sure of the reading as it is a sensitive measurement